

Partial differential equations (PDE)

A function f is a solution of the PDE when we replace ϕ with f and the left-hand side equals the right-hand side.

1.1 Laplace's equation

$$\frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} + \frac{\partial^2 \phi}{\partial z^2} = 0$$

1.2 Linear partial differential equations

All terms contain a ϕ or a partial derivative of ϕ , like $c\phi$ or $c(\text{partial derivative of } \phi)$.

1.2.1 Non-linear PDE examples

$$\frac{\partial^2 p}{\partial x^2} + \frac{\partial^2 p}{\partial x \partial y} + 3p = 0$$

$$\frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} = \sin \phi$$

1.2.2 Homogeneous linear PDE

Homogeneous linear PDEs are equal to zero ($= 0$).

1.2.3 Solving initial boundary value problems

1.2.3.1 Step 1: Method of separation

1. Let $u(x, t) = X(x)T(t)$ and get $\frac{\partial^2 X}{\partial x^2}$ and $\frac{\partial^2 T}{\partial t^2}$.

2. Substitute the equations from the previous step into the PDE.

3. Divide both sides by $X(x)T(t)$.

4. Separate $X(x)$ and $T(t)$ to the left-hand side and right-hand side respectively.

5. Obtain ordinary differential equations (ODE) LHS = γ and RHS = γ .

1.2.3.2 Step 2: Solve ordinary differential equations

Cases for the partial differential equation:

• Case 1: $\gamma = 0$

• Case 2: $\gamma > 0$, let $\gamma = p^2$, $X(x) = e^{px}$, $T(t) = e^{\lambda t}$

• Case 3: $\gamma < 0$, let $\gamma = p^2$, $X(x) = e^{px}$, $T(t) = e^{\lambda t}$

Cases for the ordinary differential equations solutions:

• Case 1: $\lambda = a, b$

$$y = Ae^{ax} + Be^{bx}$$

• Case 2: $\lambda = a$ only

$$y = (Ax + B)e^{ax}$$

• Case 3: $\lambda = \alpha \pm \beta i$

$$y = e^{\alpha x} (A \cos \beta x + B \sin \beta x)$$

Substitute the boundary conditions to find the solutions.

1.2.3.3 Step 3: Form the series solutions

1. Sum up all the solutions in step 2 to form a series solution.

2. Ensure the boundary conditions are still satisfied.

Example:

$$\phi(x, y) = \sum_{n=1}^{\infty} \sin(n\pi x) (A_n e^{n\pi y} + B_n e^{-n\pi y})$$

1.2.3.4 Step 4: Tackle initial conditions

• Use the initial conditions from the question.

• The Fourier series may be used to find unknowns.

1.3 Fourier series problem 1

$$a_0 + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{L}\right) = f(x) \text{ for } 0 < x < L$$

Where:

$$a_0 = \frac{1}{L} \int_0^L f(x) dx$$

$$a_n = \frac{2}{L} \int_0^L f(x) \cos\left(\frac{n\pi x}{L}\right) dx \text{ for } n = 1, 2, 3, \dots$$

1.4 Fourier series problem 2

$$\sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{L}\right) = f(x) \text{ for } 0 < x < L$$

Where:

$$b_n = \frac{2}{L} \int_0^L f(x) \sin\left(\frac{n\pi x}{L}\right) dx$$

2 Finite element method

1. Draw out the finite element model using the elements required.

2. Write out the element stiffness matrices.

3. Assemble the element stiffness matrices by adding them together.

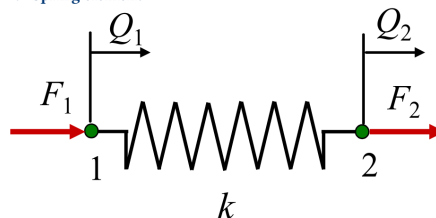
4. Apply the boundary conditions and get the system of equations and solve.

2.1 Boundary conditions

1. Zero boundary condition, $Q_n = 0$, cancel both the row and column associated with the boundary condition.

2. Non-zero boundary condition, $Q_n \neq 0$, cancel only the row associated with the boundary condition, but skip the Q_n value in the displacement vector.

2.2 Spring element

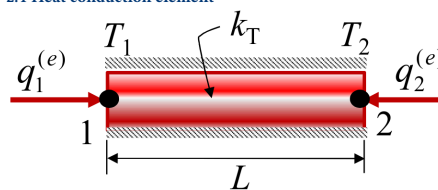


$$k \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \begin{Bmatrix} Q_1 \\ Q_2 \end{Bmatrix} = \begin{Bmatrix} F_1 \\ F_2 \end{Bmatrix}$$

2.3 Bar element (axial loading only)

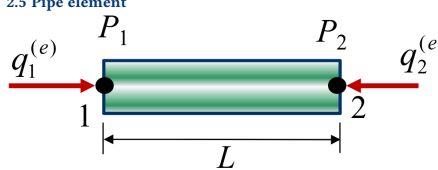
$$\frac{EA}{L} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \begin{Bmatrix} Q_1 \\ Q_2 \end{Bmatrix} = \begin{Bmatrix} F_1 \\ F_2 \end{Bmatrix}$$

2.4 Heat conduction element



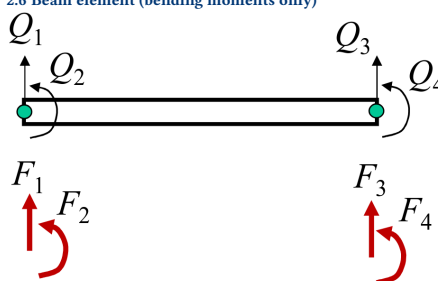
$$\frac{k_T A}{L} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \begin{Bmatrix} T_1 \\ T_2 \end{Bmatrix} = \begin{Bmatrix} q_1^{(e)} \\ q_2^{(e)} \end{Bmatrix}$$

2.5 Pipe element



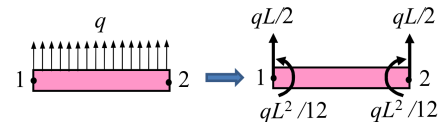
$$\frac{\pi D^4}{128 \mu L} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \begin{Bmatrix} P_1 \\ P_2 \end{Bmatrix} = \begin{Bmatrix} q_1^{(e)} \\ q_2^{(e)} \end{Bmatrix}$$

2.6 Beam element (bending moments only)



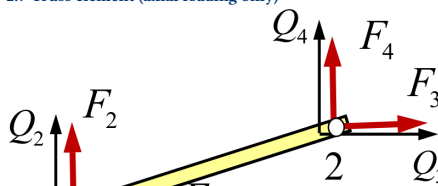
$$\frac{EI}{L^3} \begin{bmatrix} 12 & 6L & -12 & 6L \\ 6L & 4L^2 & -6L & 2L^2 \\ -12 & -6L & 12 & -6L \\ 6L & 2L^2 & -6L & 4L^2 \end{bmatrix} \begin{Bmatrix} Q_1 \\ Q_2 \\ Q_3 \\ Q_4 \end{Bmatrix} = \begin{Bmatrix} F_1 \\ F_2 \\ F_3 \\ F_4 \end{Bmatrix}$$

2.6.1 Uniformly distributed load



$$\text{For an upward distributed load: } \{f_q\} = \begin{Bmatrix} \frac{qL}{2} \\ \frac{qL^2}{12} \\ \frac{qL}{2} \\ -\frac{qL^2}{12} \end{Bmatrix}$$

2.7 Truss element (axial loading only)



$$\frac{EA}{L} \begin{bmatrix} l^2 & lm & -l^2 & -lm \\ lm & m^2 & -lm & -m^2 \\ -l^2 & -lm & l^2 & lm \\ -lm & -m^2 & lm & m^2 \end{bmatrix} \begin{Bmatrix} Q_1 \\ Q_2 \\ Q_3 \\ Q_4 \end{Bmatrix} = \begin{Bmatrix} F_1 \\ F_2 \\ F_3 \\ F_4 \end{Bmatrix}$$

• $l = \cos \theta$, $m = \sin \theta$, θ is measured from the x -axis at the reference node, node 1 in this case, anti-clockwise.

2.8 General expressions

2.8.1 Potential energy

$$U = \frac{1}{2} k (u_n - u_{n-1})^2$$

Where:

• k is the stiffness constant before the matrix in the previous section.

• u_n, u_{n-1} is the displacement.

2.8.2 Work done

$$W = F u_n, \text{ where } u_n \text{ is the displacement}$$

2.8.3 Total potential energy

$$\Pi = U - W$$

$$\Pi = \frac{1}{2} \int_V \{\epsilon\}^T \{\sigma\} dV - \int_V \{u\}^T \{b\} dV - \int_S \{u\}^T \{\bar{t}\} dS$$

Where:

• $\{u\} = \begin{Bmatrix} u \\ v \end{Bmatrix}$ is the displacement vector

• $\{b\} = \begin{Bmatrix} b_x \\ b_y \end{Bmatrix}$ is the body force vector

• $\{\bar{t}\} = \begin{Bmatrix} \bar{t}_x \\ \bar{t}_y \end{Bmatrix}$ is the surface force vector.

• $\{\sigma\} = \begin{Bmatrix} \sigma_x \\ \sigma_y \\ \tau_{xy} \end{Bmatrix}$ is the stress vector

• $\{\epsilon\} = \begin{Bmatrix} \epsilon_x \\ \epsilon_y \\ \gamma_{xy} \end{Bmatrix}$ is the strain vector.

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2.8.8 Potential energy expression

$$\Pi = \frac{1}{2} \{u^e\}^T \left(\int_V [B]^T [D] [B] dV \right) \{u^e\}$$

$$- \{u^e\}^T \left(\int_V [N]^T \{b\} dV \right)$$

$$- \{u^e\}^T \left(\int_S [N]^T \{\bar{t}\} dS \right)$$

Where:

• $[K]^e = \int_V [B]^T [D] [B] dV$ is the element stiffness matrix

• $\{f_b\}^e = \int_V [N]^T \{b\} dV$ is the element load vector due to body force

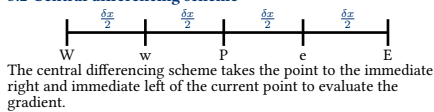
• $\{f_t\}^e = \int_S [N]^T \{\bar{t}\} dS$ is the element load vector due to surface force.

computational fluid dynamics

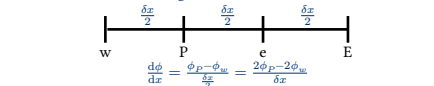
3.1 Finite difference method

The finite difference method takes the gradient by using the basic gradient formula: $\frac{d\phi}{dx} = \frac{\phi_2 - \phi_1}{x_2 - x_1}$

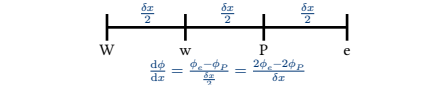
3.2 Central differencing scheme



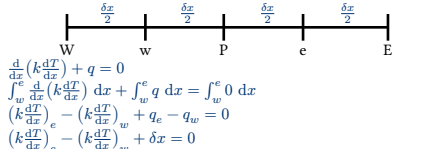
3.3 Forward differencing scheme



3.4 Backward differencing scheme



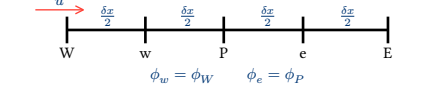
3.5 Finite volume method



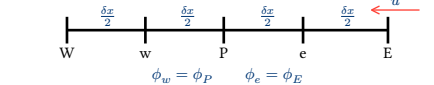
3.6 Upwind scheme

The upwind scheme is used when there is high fluid velocity (u) to get the values of ϕ_e and ϕ_w .

3.6.1 Fluid flowing to the right



3.6.2 Fluid flowing to the left



3.6.3 Peclet number

$Pe = \frac{\delta x \rho u}{\mu} < 2$ for central differencing scheme to apply

3.7 Transient response

Future T: $T(\Delta t) = T^{(1)}$, Past T: $T(0) = T^{(0)}$

The formulas below are only used when the equation cannot be directly integrated to remove the derivative.

- Implicit scheme for time:

$$\int_0^{\Delta t} f(t) dt = f(\Delta t) \Delta t$$

- Explicit scheme for time:

$$\int_0^{\Delta t} f(t) dt = f(0) \Delta t$$

Stability criterion:

$$\Delta t < \rho \frac{(\delta x)^2}{2k}$$

- Estimation of space:

$$\int_a^b f(x) dx = f\left(\frac{a+b}{2}\right)(b-a)$$

3.8 Iterative techniques

$$\begin{bmatrix} 2 & 1 & 1 \\ -1 & 3 & -1 \\ 1 & -1 & 2 \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \end{Bmatrix} = \begin{Bmatrix} 7 \\ 2 \\ 5 \end{Bmatrix}$$

Equation form:

$$2x_1 + x_2 + x_3 = 7$$

$$-x_1 + 3x_2 - x_3 = 2$$

$$x_1 - x_2 + 2x_3 = 5$$

Always rearrange the first equation to place x_1 on the left-hand side, the second equation to get x_2 on the left-hand side, and so on:

$$x_1 = \frac{7 - x_2 - x_3}{2}$$

$$x_2 = \frac{2 + x_1 + x_3}{3}$$

$$x_3 = \frac{5 - x_1 + x_2}{2}$$

3.8.1 Scarborough criterion

- For every row in the matrix, look at the diagonal value in the matrix and take the absolute value. This term is denoted as $|a_P|$.

- Then, sum up the absolute values of the values beside that diagonal value. This term is denoted as $\sum |a_{nb}|$.

- The Scarborough criterion is:

$$\sum_{n \neq P} \frac{|a_{nP}|}{|a_{PP}|} \leq 1 \text{ for every row in the matrix, and } < 1 \text{ for at least one row in the matrix}$$

3.8.2 Jacobi method

- Guess values on the right-hand side, like setting everything to 0 for example.

- New values can be calculated on the left-hand side.

- Substitute the new values into the right-hand side and evaluate the left-hand side again.

- Repeat the process until there is no more change in the solution.

3.8.3 Gauss-Seidel method

This method is similar to the Jacobi method, but instead of solving all the simultaneous equations at the same time to get the value of each unknown, solve each equation individually, then use the new value when solving the next equation for the next unknown.

3.8.3.1 Example

$$\begin{aligned} x_1 &= \frac{7 - x_2 - x_3}{2} & x_1^{(1)} &= \frac{7 - x_2^{(0)} - x_3^{(0)}}{2} = \frac{7 - 0 - 0}{2} = \frac{7}{2} \\ x_2 &= \frac{2 + x_1 + x_3}{3} & x_2^{(1)} &= \frac{2 + x_1^{(1)} + x_3^{(0)}}{3} = \frac{2 + 3.5 + 0}{3} = \frac{11}{6} \\ x_3 &= \frac{5 - x_1 + x_2}{2} & x_3^{(1)} &= \frac{5 - x_1^{(1)} + x_2^{(1)}}{2} = \frac{5 - 3.5 + 1.8333}{2} = \frac{3.3333}{2} = 1.6667 \end{aligned}$$

Where:

- $x^{(0)}$ denotes the start value, or iteration 0

- $x^{(1)}$ denotes iteration 1

- $x^{(n)}$ denotes iteration n

4 Maths

4.1 Derivatives

Chain rule:

$$\frac{dz}{dx} = \frac{dz}{dy} \cdot \frac{dy}{dx}$$

Product rule:

$$\frac{d}{dx} (u \cdot v) = \frac{du}{dx} \cdot v + u \cdot \frac{dv}{dx}$$

Quotient rule:

$$\frac{d}{dx} \left(\frac{f(x)}{g(x)} \right) = \frac{f'(x)g(x) - f(x)g'(x)}{g(x)^2}$$

Standard derivatives:

$$\frac{d}{dx} (\sin x) = \cos x$$

$$\frac{d}{dx} (\cos x) = -\sin x$$

$$\frac{d}{dx} (\arcsin x) = \frac{1}{\sqrt{1-x^2}}$$

$$\frac{d}{dx} (\arccos x) = -\frac{1}{\sqrt{1-x^2}}$$

$$\frac{d}{dx} (\arctan x) = \frac{1}{1+x^2}$$

$$\frac{d}{dx} (\csc x) = -\csc x \cot x$$

$$\frac{d}{dx} (\sec x) = \sec x \tan x$$

4.2 Integrals

$$\int \frac{1}{x} = \ln|x|$$

$$\int \sin x dx = -\cos x$$

$$\int \cos x dx = \sin x$$

$$\int \frac{1}{x^2 + a^2} dx = \frac{1}{a} \arctan\left(\frac{x}{a}\right)$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \arcsin\left(\frac{x}{a}\right)$$

$$\int \frac{1}{x^2 - a^2} dx = \frac{1}{2a} \ln \left| \frac{x-a}{x+a} \right|$$

$$\int \frac{1}{a^2 - x^2} dx = \frac{1}{2a} \ln \left| \frac{a+x}{a-x} \right|$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left| \sqrt{x^2 - a^2} + x \right|$$

$$\int \tan x dx = \ln|\sec x|$$

$$\int \cot x dx = \ln|\sin x|$$

$$\int \csc x dx = -\ln|\csc x + \cot x|$$

$$\int \sec x dx = -\ln|\sec x + \tan x|$$

Integration by parts:

$$\int u dv = uv - \int v du$$

Preference for u :

- L: log or ln.
- I: Inverse trigonometric functions.
- A: Algebraic functions.
- T: Trigonometric functions.
- E: Exponential functions.

4.3 Trigonometric identities

Quotient identities:

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

$$\cot \theta = \frac{\cos \theta}{\sin \theta}$$

Reciprocal identities:

$$\sin \theta = \frac{1}{\csc \theta}$$

$$\csc \theta = \frac{1}{\sin \theta}$$

$$\cos \theta = \frac{1}{\sec \theta}$$

$$\sec \theta = \frac{1}{\cos \theta}$$

$$\tan \theta = \frac{1}{\cot \theta}$$

$$\cot \theta = \frac{1}{\tan \theta}$$

Pythagorean identities:

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\sec^2 \theta - \tan^2 \theta = 1$$

$$\csc^2 \theta - \cot^2 \theta = 1$$

Even/odd identities:

$$\sin(-\theta) = -\sin \theta$$

$$\cos(-\theta) = \cos \theta$$

$$\tan(-\theta) = -\tan \theta$$

$$\cot(-\theta) = -\cot \theta$$

$$\csc(-\theta) = -\csc \theta$$

$$\sec(-\theta) = \sec \theta$$

Co-function identities:

$$\sin\left(\frac{\pi}{2} - \theta\right) = \cos \theta$$

$$\cos\left(\frac{\pi}{2} - \theta\right) = \sin \theta$$

$$\tan\left(\frac{\pi}{2} - \theta\right) = \cot \theta$$

$$\cot\left(\frac{\pi}{2} - \theta\right) = \tan \theta$$

$$\csc\left(\frac{\pi}{2} - \theta\right) = \sec \theta$$

$$\sec\left(\frac{\pi}{2} - \theta\right) = \csc \theta$$

$$\frac{\pi}{2} \text{ radians} = 90^\circ$$

Sum/difference identities:

$$\sin(\theta \pm \phi) = \sin \theta \cos \phi \pm \cos \theta \sin \phi$$

$$\cos(\theta \pm \phi) = \cos \theta \cos \phi \mp \sin \theta \sin \phi$$

$$\tan(\theta \pm \phi) = \frac{\tan \theta \pm \tan \phi}{1 \mp \tan \theta \tan \phi}$$

Double angle identities:

$$\sin(2\theta) = 2 \sin \theta \cos \theta$$

$$\cos(2\theta) = \cos^2 \theta - \sin^2 \theta$$

$$\cos(2\theta) = 2 \cos^2 \theta - 1$$

$$\cos(2\theta) = 1 - 2 \sin^2 \theta$$

$$\tan(2\theta) = \frac{2 \tan \theta}{1 - \tan^2 \theta}$$

Half angle identities:

$$\sin^2 \theta = \frac{1 - \cos(2\theta)}{2}$$

$$\cos^2 \theta = \frac{1 + \cos(2\theta)}{2}$$

$$\tan^2 \theta = \frac{1 - \cos(2\theta)}{1 + \cos(2\theta)}$$

Sum to product of 2 angles:

$$\sin \theta + \sin \phi = 2 \sin\left(\frac{\theta + \phi}{2}\right) \cos\left(\frac{\theta - \phi}{2}\right)$$

$$\sin \theta - \sin \phi = 2 \cos\left(\frac{\theta + \phi}{2}\right) \sin\left(\frac{\theta - \phi}{2}\right)$$

$$\cos \theta + \cos \phi = 2 \cos\left(\frac{\theta + \phi}{2}\right) \cos\left(\frac{\theta - \phi}{2}\right)$$

$$\cos \theta - \cos \phi = -2 \sin\left(\frac{\theta + \phi}{2}\right) \sin\left(\frac{\theta - \phi}{2}\right)$$

Product to sum of 2 angles:

$$\sin \theta \sin \phi = \frac{\cos(\theta - \phi) - \cos(\theta + \phi)}{2}$$

$$\cos \theta \cos \phi = \frac{\cos(\theta - \phi) + \cos(\theta + \phi)}{2}$$

$$\sin \theta \cos \phi = \frac{\sin(\theta + \phi) + \sin(\theta - \phi)}{2}$$

$$\cos \theta \sin \phi = \frac{\sin(\theta + \phi) - \sin(\theta - \phi)}{2}$$

Law of sines:

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

Law of cosines:

$$a^2 = b^2 + c^2 - 2bc \cos A$$

Area of a triangle:

$$A = \frac{1}{2} ab \sin C$$